

Holiday Itinerary Data Platform

- > From DATAtourisme raw data to an interactive Streamlit itinerary dashboard
- > A complete ETL + API + visualization stack

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The problem we solved

Why

Tourism data is rich, but raw feeds are not directly usable for itinerary planning

Raw feed

Nested JSON, inconsistent field shapes, multilingual values, changing source files

Planning need

Users need city-level exploration, categories, maps, and day-by-day suggestions

Engineering need

Repeatable ingestion, incremental processing, and clear deployment packaging

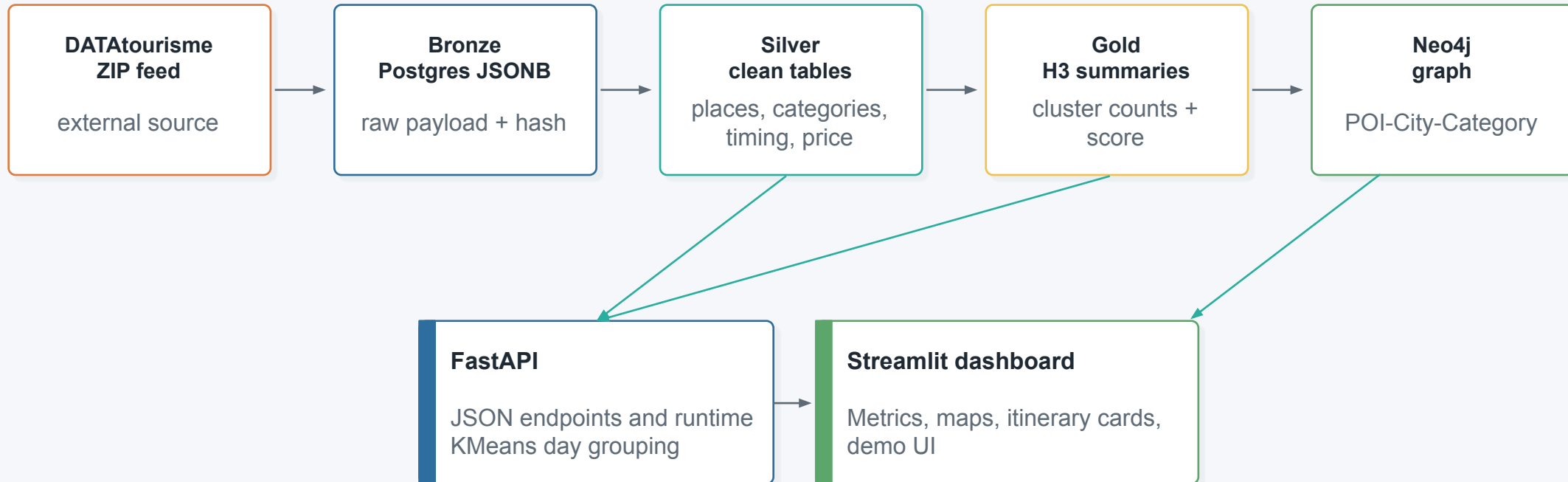
Our outcome

A Dockerized ETL + API + dashboard stack that converts raw POIs into a demoable itinerary application.

End-to-end architecture

Architecture

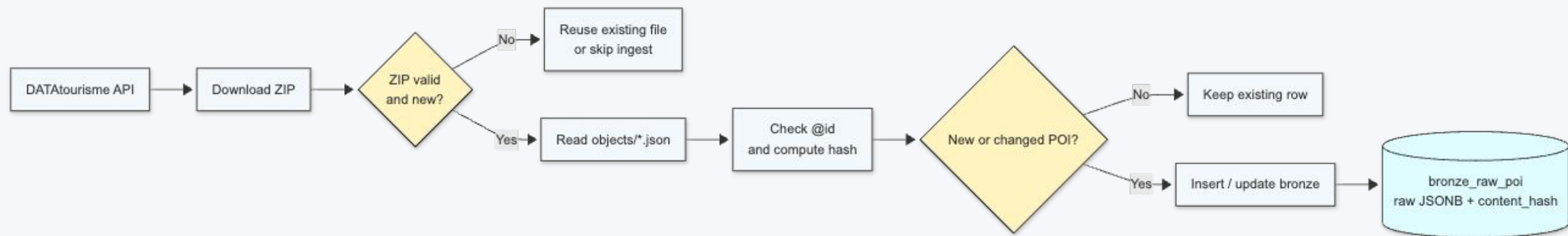
Two clean lanes: ETL builds trusted data; app services turn it into an itinerary experience



End-to-end architecture

API ingestion

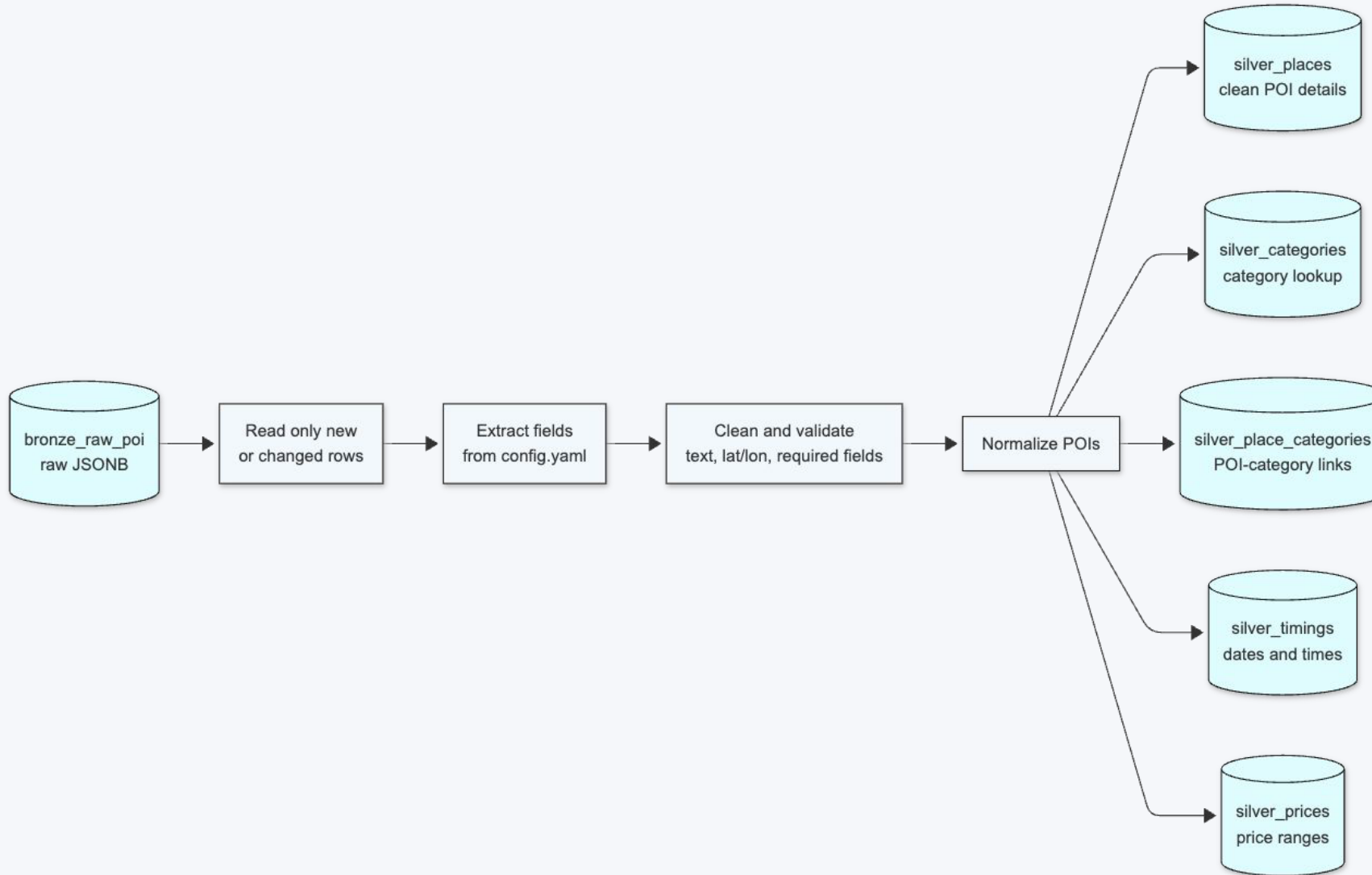
Architecture



End-to-end architecture

Architecture

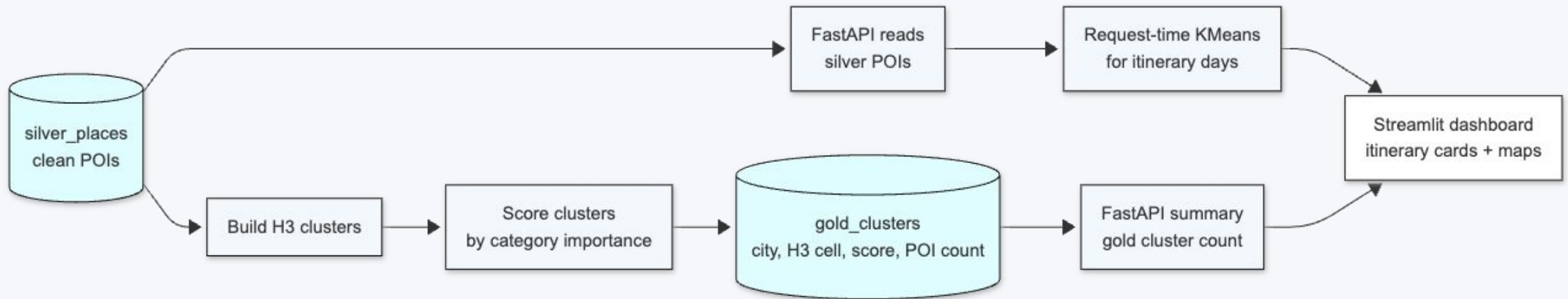
Bronze to silver database



End-to-end architecture

Architecture

Silver to gold database



Implemented stack and deferred options

Stack area

We used proven tools for local reproducibility and explainability; heavier options stayed out of scope for scale, setup, or time reasons

Orchestration	Python CLI, cron, and Docker Compose for repeatable ETL runs
Storage	Postgres JSONB for bronze, relational silver tables, and H3 gold summaries
Transformation + ML	SQL, Pandas chunks for cleaning; scikit-learn, KMeans for request-time day grouping
Graph layer	Neo4j for POI-City-Category relationships and related-place suggestions
API + UI	FastAPI as the JSON boundary; Streamlit for the interactive itinerary demo
Delivery	GitHub Actions checks plus Docker Hub image publishing for reusable releases

Itinerary generation: request-time intelligence

Algorithm

KMeans creates day areas; interests guide selection without over-filtering

1. User input

city, days, places/day,
interests

2. Silver POIs

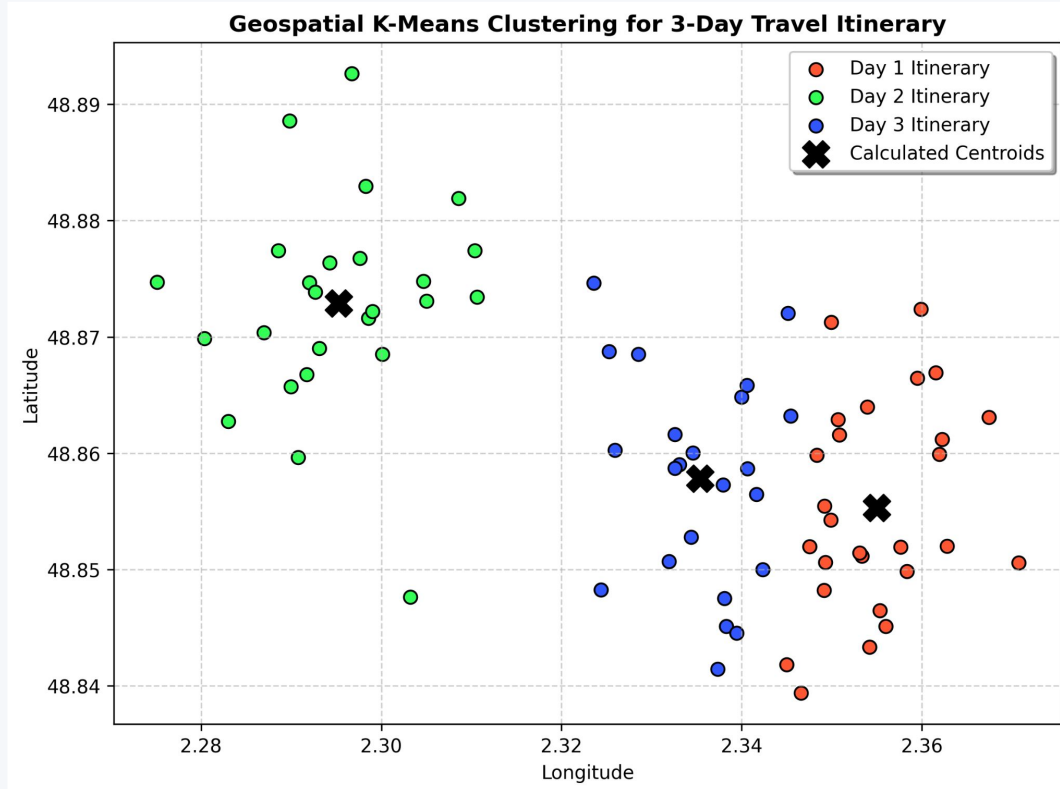
read cleaned lat/lon +
categories

3. KMeans

N clusters =
requested days

4. Day plan

nearest to center +
interest preference



Interests are preferences, not hard filters

Example: selecting Beach nudges the planner to include beach-related POIs, but each day is still filled with nearby varied stops.

Beach

Museum

Food

default



GET	/health	Health	^
Parameters			Try it out
No parameters			
Responses			
Code	Description		Links
200	Successful Response		No links
	Media type		
	application/json		
	Controls Accept header.		
	Example Value Schema		
	"string"		
GET	/summary	Summary	^
GET	/cities	Cities	^
GET	/categories	Categories	^
GET	/places	Places	^
POST	/generate-itinerary	Generate Itinerary	^

Streamlit demo walkthrough

Demo

The dashboard turns backend services into an instructor-friendly product demo

localhost:8501

Holiday Itinerary Dashboard

Connected to <http://api:8000>

Places

575,312

Cities

27,837

Categories

422

Gold clusters

65,997

Plan

City

Arles

Days

2

Places per day

5

Generate itinerary

Explore

Interests

AdventurePark ArcheologicalSite

Places to show

50

Itinerary

Places

Cities

Day 1

09:00 - 11:00 | Aire de pique-nique de Peaudure

ConvenientService, PicnicArea, PlaceOfInterest

RD 36

Related: No graph match yet

11:30 - 13:30 | Balade en 4x4 à la manade Jacques Bon

EntertainmentAndEvent, PointOfInterest, Product

Le Sambuc

Related: No graph match yet

14:00 - 16:00 | Aire de pique-nique de Louisianne

ConvenientService, PicnicArea, PlaceOfInterest

D36

Related: No graph match yet

16:30 - 18:30 | Aire de pique-nique de Constantin

ConvenientService, PicnicArea, PlaceOfInterest

aire de constantin

Related: No graph match yet

Streamlit demo walkthrough

Demo

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localhost:8501

related: no graph match yet

14:00 - 16:00 | Arles Plaza

Accommodation, Hotel, HotelTrade

45 Avenue Sadi Carnot

Related: No graph match yet

16:30 - 18:30 | Découverte de la Camargue en 4X4 avec Camargue Alpilles Safari

EntertainmentAndEvent, PointOfInterest, Product

1 Boulevard Emile Fassin

Related: No graph match yet

19:00 - 21:00 | Camargue Alpilles Safari

ActivityProvider, PlaceOfInterest, PointOfInterest

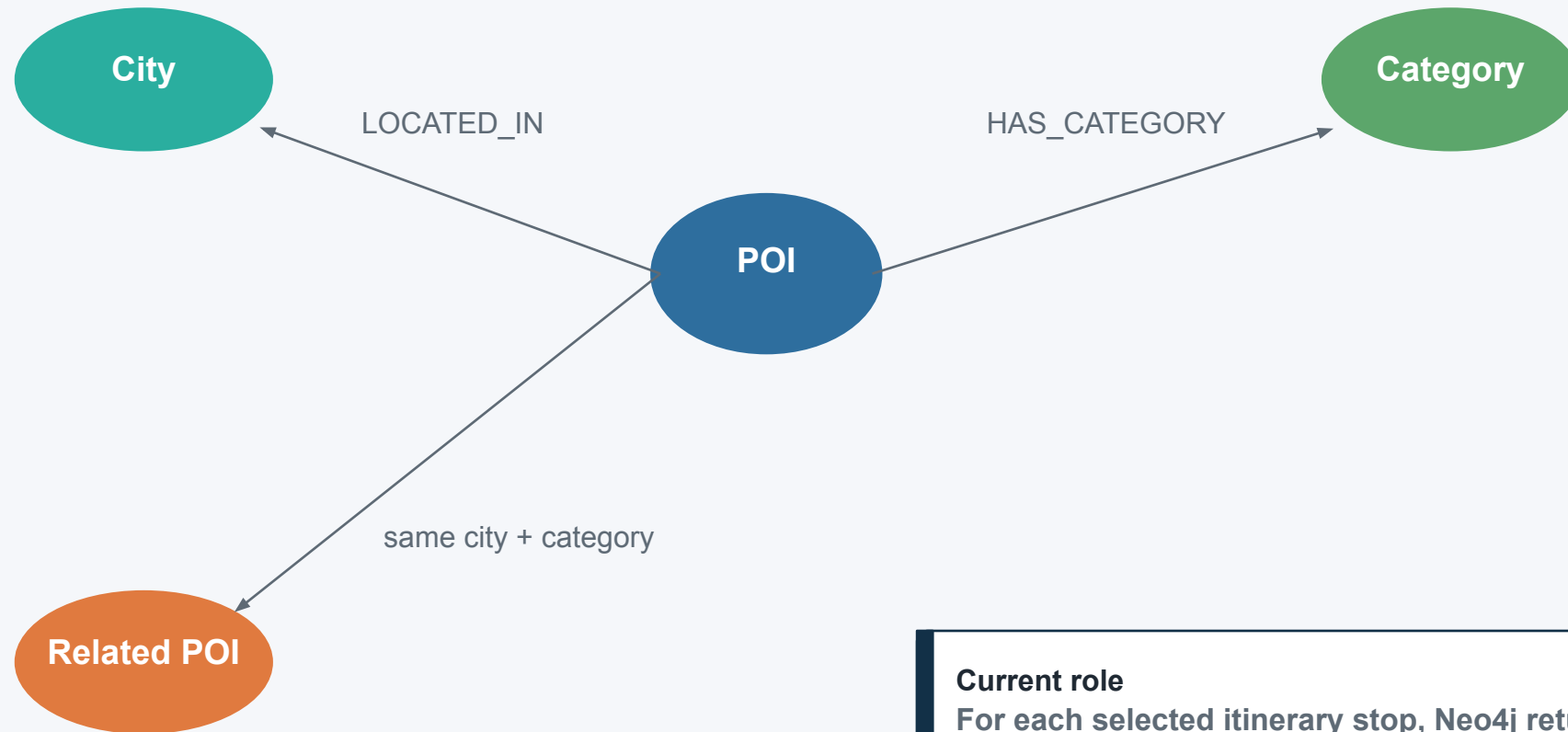
1, rue Emile Fassin

Related: No graph match yet

Where Neo4j fits

Graph

Neo4j adds relationship-aware suggestions; it does not choose the map or the day clusters



Current role

For each selected itinerary stop, Neo4j returns related POIs to show under the card.

Streamlit demo walkthrough

Demo

The dashboard turns backend services into an instructor-friendly product demo

localhost:8501

Plan

City

Paris

Days

1

Places per day

1

Generate itinerary

Explore

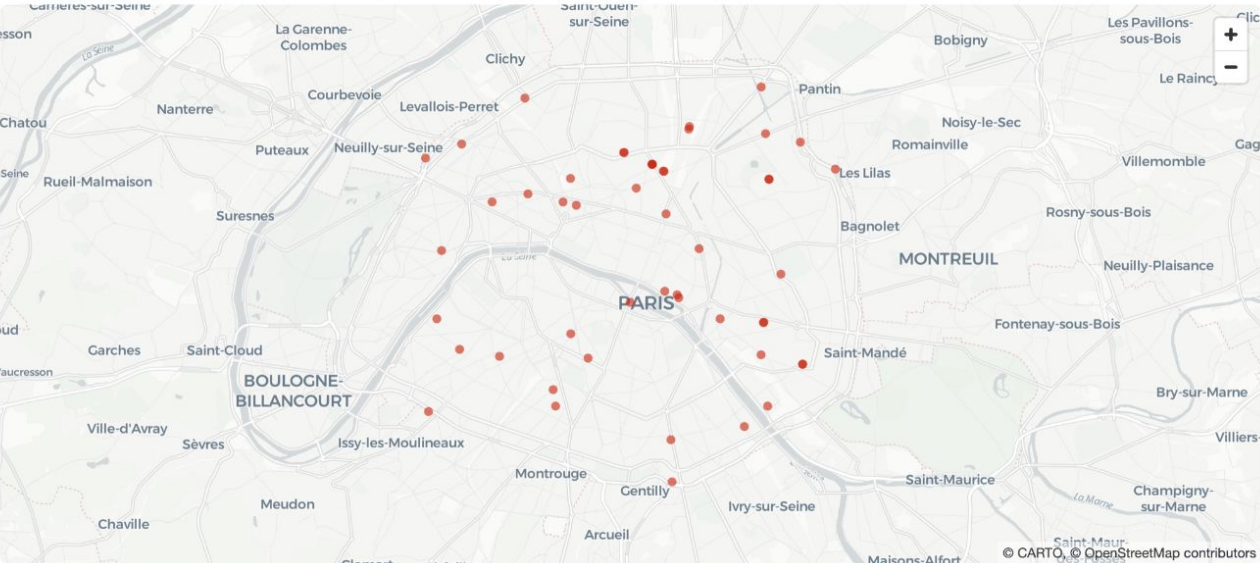
Interests

Accommodation

Places to show

50

Itinerary Places Cities

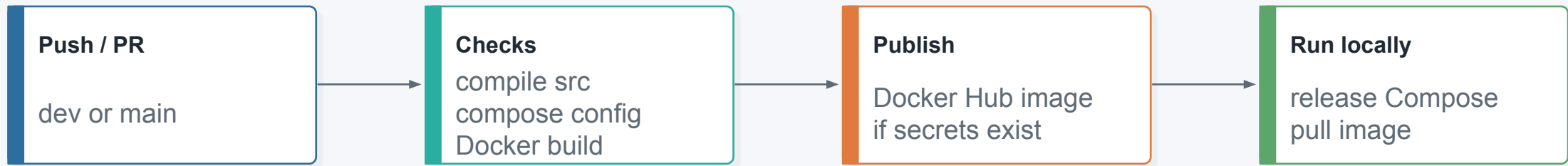


name	city	address	categories	website
1K Paris – Machefert Grou	Paris	13 boulevard du Temple	Accommodation, Hotel, HotelTrade, PlaceOfInterest, PointOfInterest, schem	https://www.1k-paris.com/
25hours Terminus Nord	Paris	12 boulevard de Denain	Accommodation, Hotel, HotelTrade, PlaceOfInterest, PointOfInterest, schem	https://www.25hours-hotels.com/fr/hc
25hours Terminus Nord	Paris	12 boulevard de Denain	Accommodation, Hotel, HotelTrade, PlaceOfInterest, PointOfInterest, schem	https://www.25hours-hotels.com/en/h
3 Ducks Boutique Hostel	Paris	6 place Etienne Pernet	Accommodation, CollectiveAccommodation, PlaceOfInterest, PointOfInteres	http://www.3ducks.fr/
AC Hotel Paris Porte Maill	Paris	6 rue Gustave Charpentier	Accommodation, Hotel, HotelTrade, PlaceOfInterest, PointOfInterest, schem	http://www.achotelparismaillot.com/
Adagio access Paris Bastil	Paris	11 rue Moreau	Accommodation, CollectiveAccommodation, HolidayResort, PlaceOfInterest,	http://www.le300.fr/
Adagio access Paris La Vill	Paris	28 bis avenue Corentin Cariou	Accommodation, CollectiveAccommodation, HolidayResort, PlaceOfInterest,	https://www.adagio-city.com/fr/hotel-
Adagio access Paris Philli	Paris	12 avenue Pierre Bayle	Accommodation, CollectiveAccommodation, HolidayResort, PlaceOfInterest,	http://www.adagio-city.com/

CI/CD and distribution

Delivery

GitHub Actions checks the project and publishes a reusable Docker image



Docker Hub is not hosting

It stores a versioned app image. A machine or server still pulls and runs the containers.

Distribution method

Users can run ``distribution/compose.release.yml`` with an ``.env`` and the published image.

Engineering decisions and known limits

Evaluation

We optimized for reproducibility and clear boundaries; next steps improve route intelligence

Good decisions

- Incremental ETL with hashes and sync timestamps
- Silver layer as trusted source of detailed POIs
- FastAPI boundary between UI and data services
- Docker Compose for reproducible demos

Next improvements

- Real route optimization or travel-time API
- Better ranking from ratings, opening hours, prices
- Hosted demo environment, not only local Compose
- Monitoring ETL status in the dashboard

What we delivered

[Close](#)

A working, explainable data platform with a demoable itinerary application

4

data layers

2

app services

1

Dockerized stack

CI

validated delivery

Tourism data is rich — but raw feeds are not directly usable for itinerary planning. We built a pipeline that bridges the gap between messy source data and a clean, demoable product.

Raw Feed Reality

- Nested JSON with inconsistent field shapes
- Multilingual values across sources
- Changing source file structures

What Users & Engineering Needed

- City-level exploration with categories and maps
- Day-by-day itinerary suggestions
- Repeatable ingestion and incremental processing
- Clear, reproducible deployment packaging

✔ **Our outcome:** A Dockerized ETL + API + dashboard stack that converts raw POIs into a demoable itinerary application.